การสื่อสารเป็นสิ่งสำคัญในการดำรงชีวิต

SIGN LANGUAGE RECOGNITION

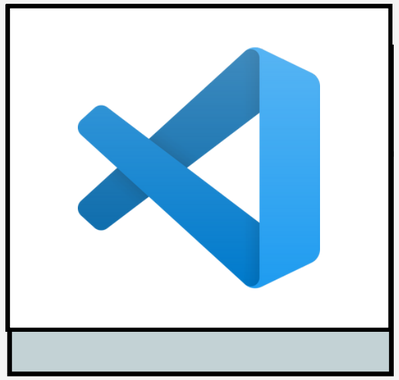


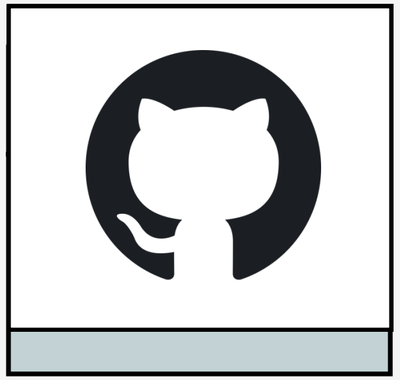
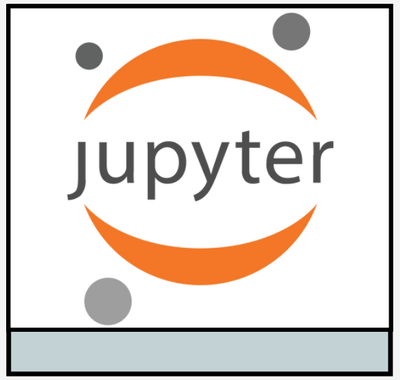
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SIGN LANGUAGE RECOGNITION



โปรแกรมและเครื่องมือ





โปรแกรมและเครื่องมือที่ใช้ในการเขียนเว็บไซต์ สื่อสารต่างๆ เพื่อใช้ในการทำงานร่วมกัน และเก็บฐานข้อมูล

การออกแบบและการพัฒนา

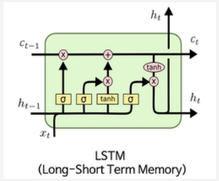


MODEL

Numpy

CV2

Tensorflow



LSTM

Mediapipe

WEBS

Mango DB
Atlas

Django
MVT

HTML ,
CSS ,
Javascript

PREPARING DATASET



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DATASET

17 ท่า ได้แก่ รัก ฉัน เธอ สวัสดี สบายดี ชอบ ดีใจ เสียใจ โกรธ ร้องไห้ กลัว อิ่ม ยาก ง่าย ลืม คิดถึง ยืน

โดยที่ แต่ละท่า มี 100 VIDEO

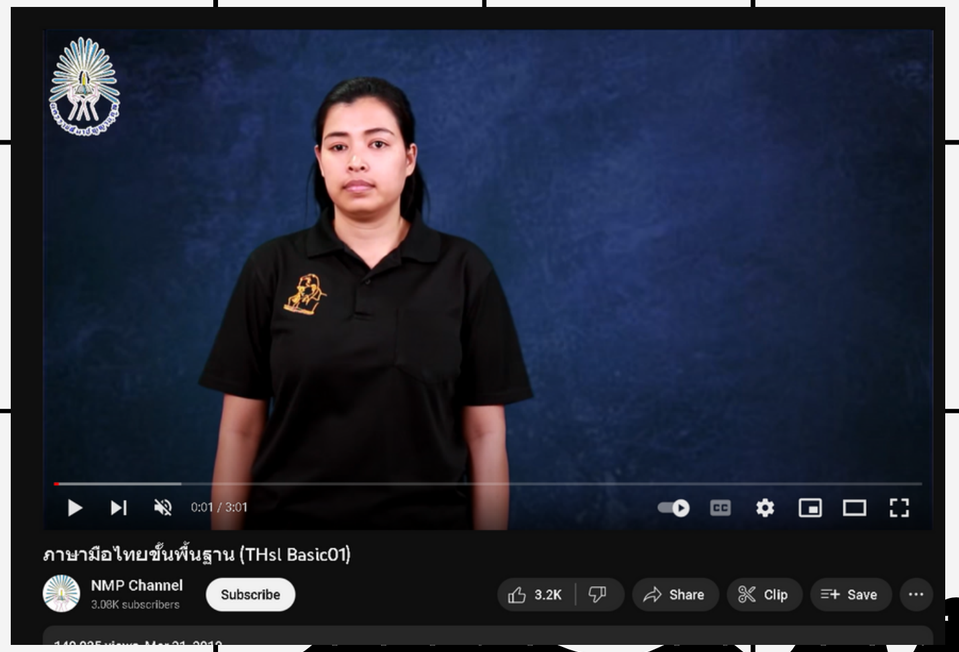
1 VIDEO มี 30 FRAMES

NMP Channel

@nmpchannel2632 3.08K subscribers 82 videos

More about this channel >

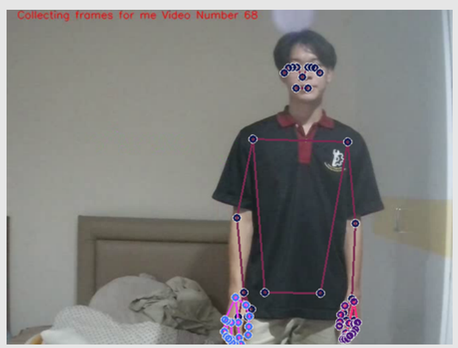


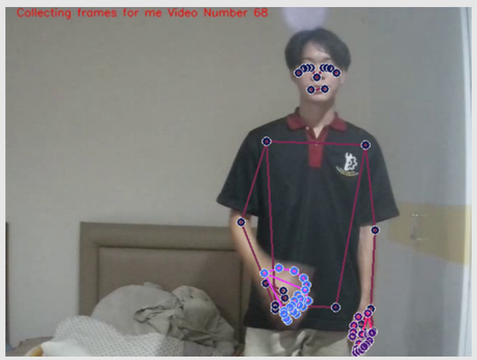


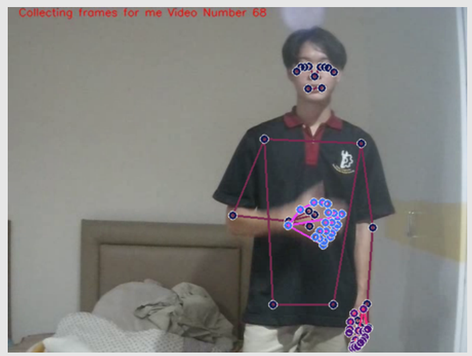
640X480 FPS 25

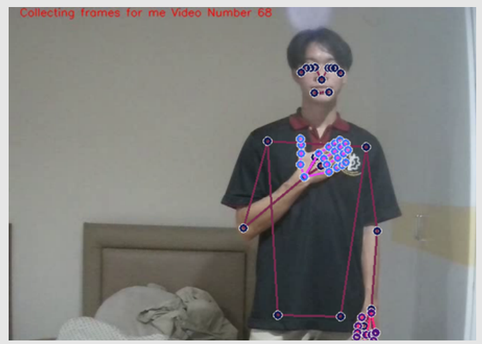


0 . . . 12 . . . 20 . . . 29









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  7.78179884e-01  8.70838881e-01  1.26267309e-07  7.58489192e-01
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```





.NPY

SPLIT



- Train : 60%
- Val : 20%
- Test : 20%

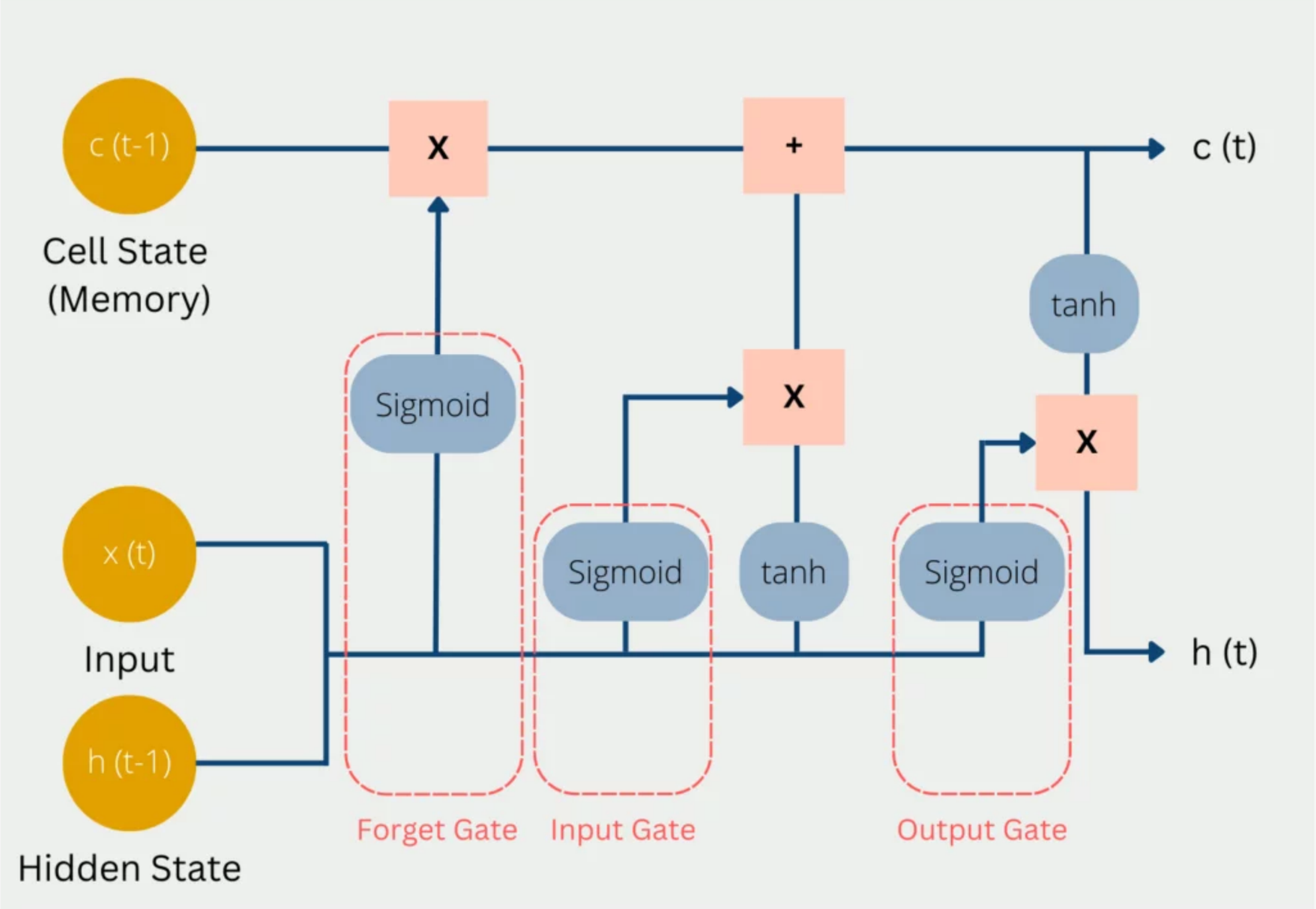
AUGMENTATION

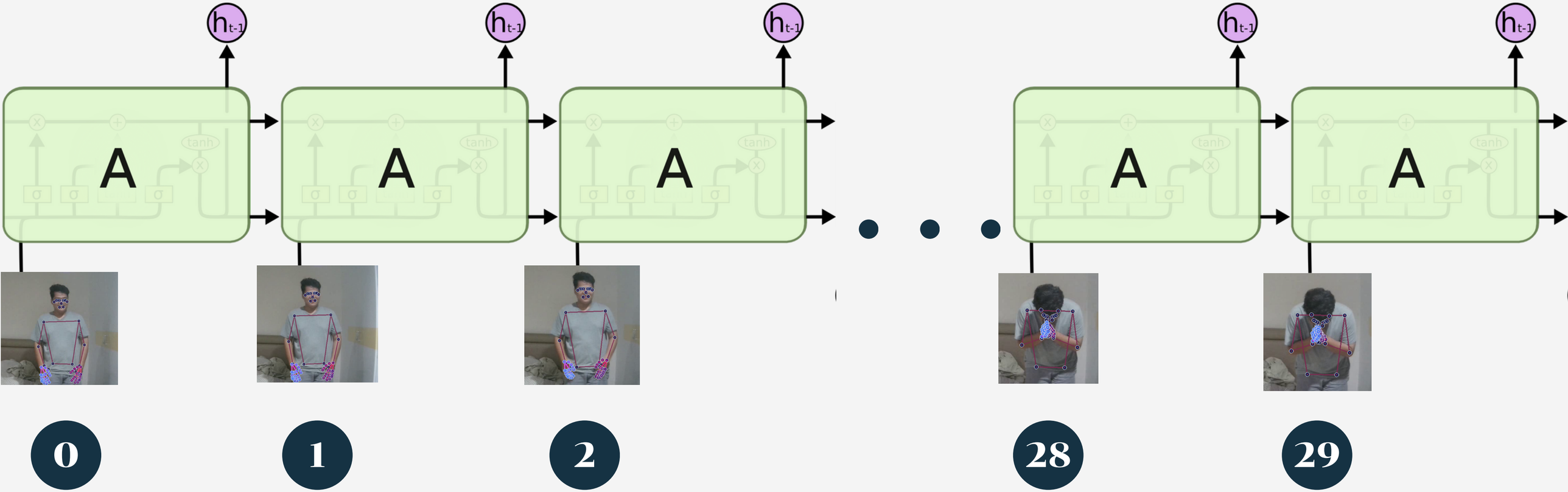


- ZOOM (1,1.5)
- ROTATION (-20,20)

MODEL?

LSTM MODEL


c (t-1)
Cell State
(Memory)
x (t)
Input
h (t-1)
Hidden State
x
+
c (t)
Sigmoid
x
tanh
x
Sigmoid
tanh
Sigmoid
h (t)
Forget Gate
Input Gate
Output Gate


h t-1
A
tanh
σ
0
1
2
28
29

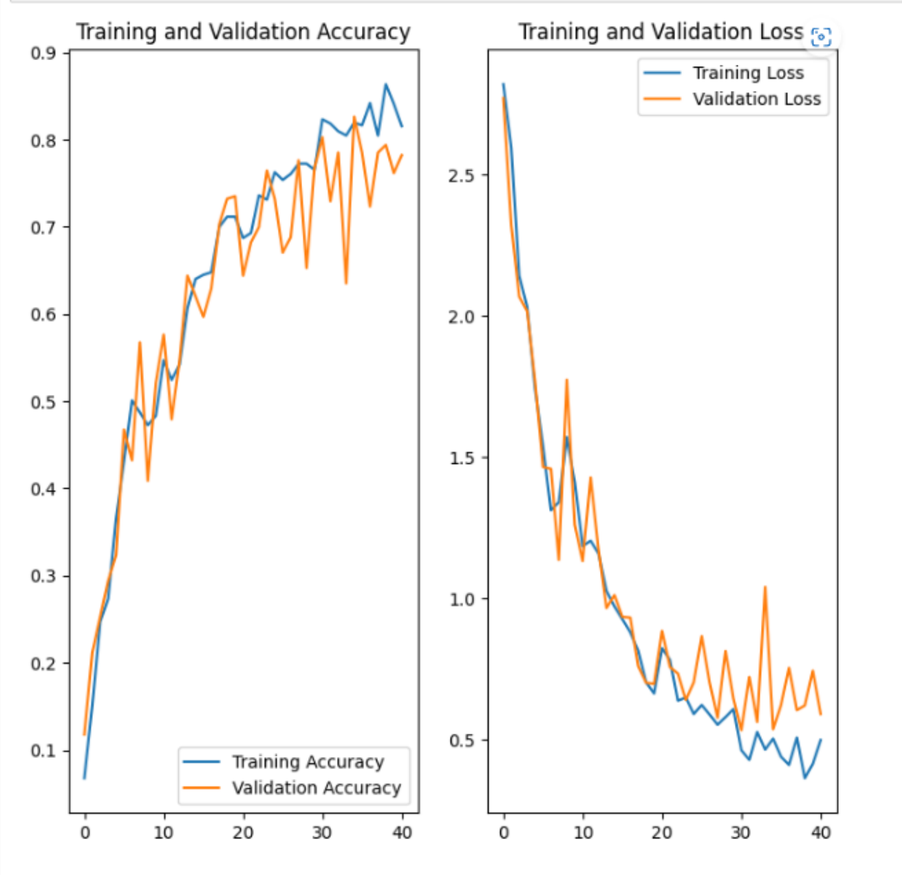
```
model = Sequential()
model.add(LSTM(64, return_sequences=True, activation='relu', input_shape=(30,258)))
model.add(LSTM(128, return_sequences=True, activation='relu'))
model.add(LSTM(64, return_sequences=False, activation='relu'))
model.add(Dense(64, activation='relu'))
model.add(Dense(32, activation='relu'))
model.add(Dense(actions.shape[0], activation='softmax'))
```

batch_size=32, learning_ rate=0.001

CALLBACKS=[EARLY_STOP,CHECK_POINT]

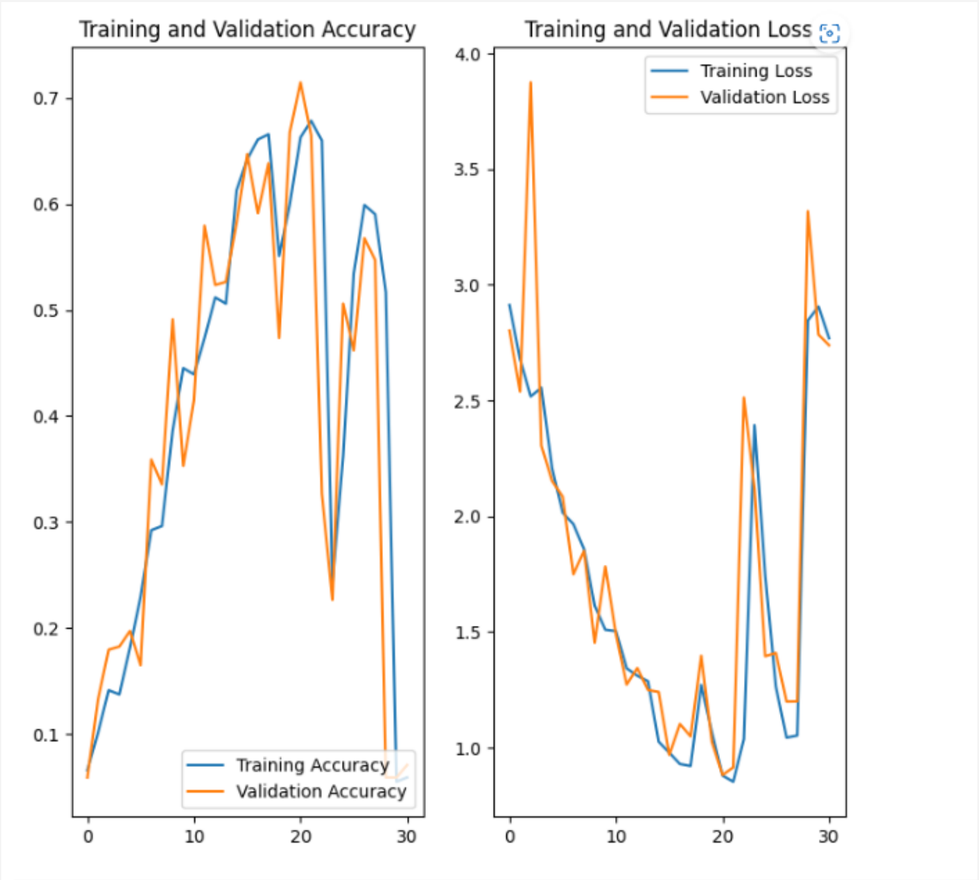


NO AUGMENTATION
ท่าละ 100, ทั้งหมด 1700



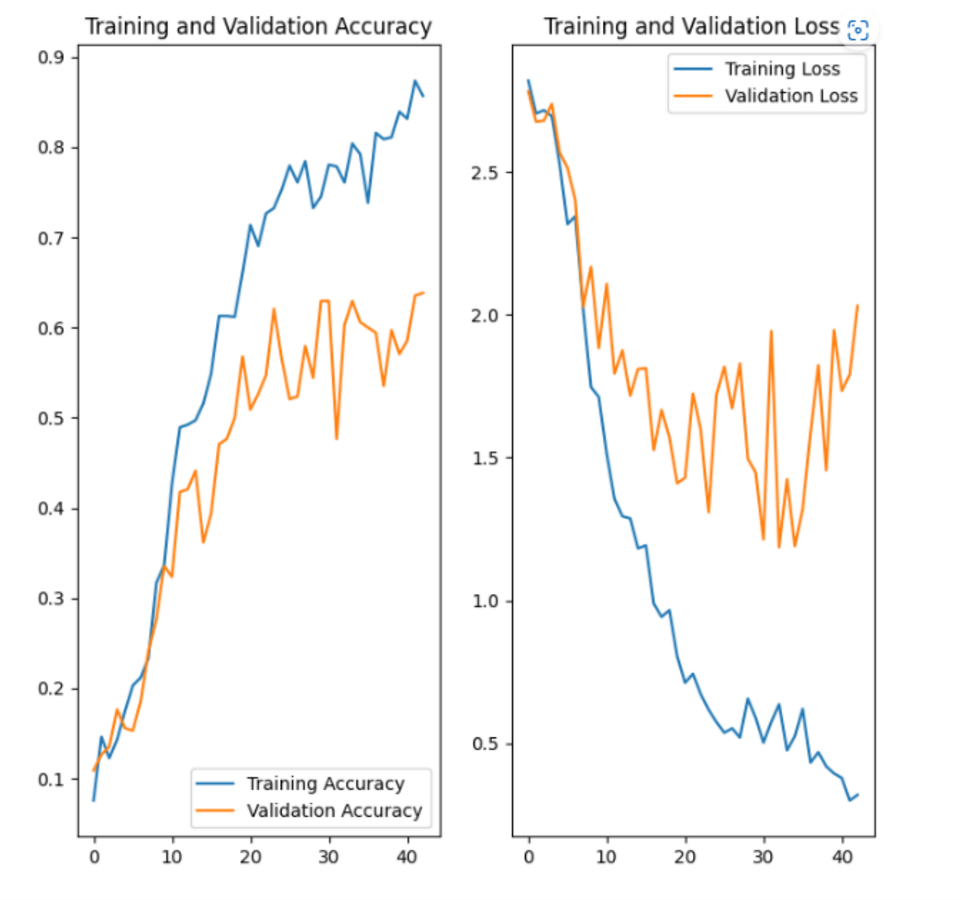
ACCURACY: 0.7618 , LOSS: 0.5604

3 AUGMENTATION(ZOOM)
ท่าละ 400 , ทั้งหมด 6800



ACCURACY: 0.6824 , LOSS: 0.9195

5 AUGMENTATION (ZOOM+AUGMENTATION)
ท่าละ 600 , ทั้งหมด 10200

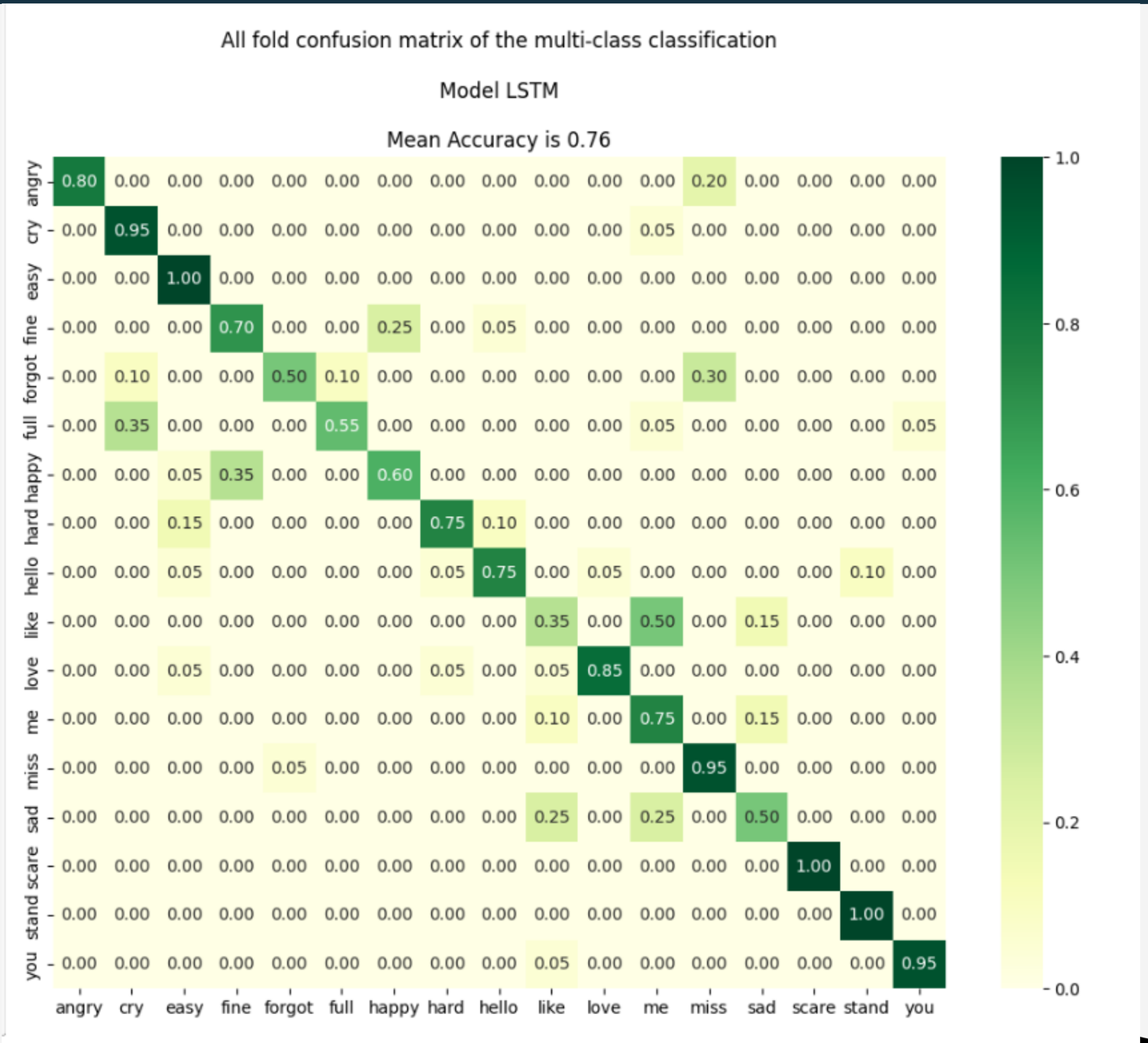


ACCURACY: 0.600 , LOSS: 1.4353

10 AUGMENTATION (ZOOM+AUGMENTATION)
ท่าละ 1000 , ทั้งหมด 17000



ACCURACY: 0.4206 , LOSS: 1.9696

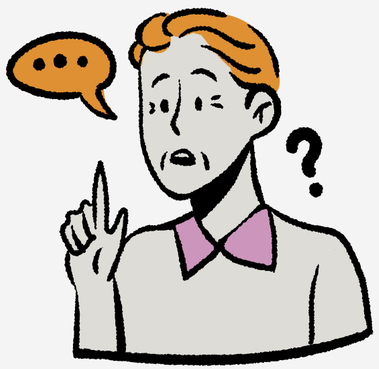

All fold confusion matrix of the multi-class classification
Model LSTM
Mean Accuracy is 0.76
angry 0.80 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.20 0.00 0.00 0.00 0.00
cry 0.00 0.95 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.05 0.00 0.00 0.00 0.00 0.00
easy 0.00 0.00 1.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
fine 0.00 0.00 0.00 0.70 0.00 0.00 0.25 0.00 0.05 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
forgot 0.00 0.10 0.00 0.00 0.50 0.10 0.00 0.00 0.00 0.00 0.00 0.00 0.30 0.00 0.00 0.00 0.00
full 0.00 0.35 0.00 0.00 0.00 0.55 0.00 0.00 0.00 0.00 0.00 0.05 0.00 0.00 0.00 0.00 0.05
happy 0.00 0.00 0.05 0.35 0.00 0.00 0.60 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
hard 0.00 0.00 0.15 0.00 0.00 0.00 0.00 0.75 0.10 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
hello 0.00 0.00 0.05 0.00 0.00 0.00 0.00 0.05 0.75 0.00 0.05 0.00 0.00 0.00 0.00 0.10 0.00
like 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.35 0.00 0.50 0.00 0.15 0.00 0.00 0.00
love 0.00 0.00 0.05 0.00 0.00 0.00 0.00 0.05 0.00 0.05 0.85 0.00 0.00 0.00 0.00 0.00 0.00
me 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.10 0.00 0.75 0.00 0.15 0.00 0.00 0.00
miss 0.00 0.00 0.00 0.00 0.05 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.95 0.00 0.00 0.00 0.00
sad 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.25 0.00 0.25 0.00 0.50 0.00 0.00 0.00
scare 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 1.00 0.00 0.00
stand 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 1.00 0.00
you 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.05 0.00 0.00 0.00 0.00 0.00 0.00 0.95
angry cry easy fine forgot full happy hard hello like love me miss sad scare stand you
1.0
0.8
0.6
0.4
0.2
0.0

	precision	recall	f1-score	support
angry	1.00	0.80	0.89	20
cry	0.68	0.95	0.79	20
easy	0.77	1.00	0.87	20
fine	0.67	0.70	0.68	20
forgot	0.91	0.50	0.65	20
full	0.85	0.55	0.67	20
happy	0.71	0.60	0.65	20
hard	0.88	0.75	0.81	20
hello	0.83	0.75	0.79	20
like	0.44	0.35	0.39	20
love	0.94	0.85	0.89	20
me	0.47	0.75	0.58	20
miss	0.66	0.95	0.78	20
sad	0.62	0.50	0.56	20
scare	1.00	1.00	1.00	20
stand	0.91	1.00	0.95	20
you	0.95	0.95	0.95	20
accuracy			0.76	340
macro avg	0.78	0.76	0.76	340
weighted avg	0.78	0.76	0.76	340



ปัญหาที่พบ&วิธีแก้ไข

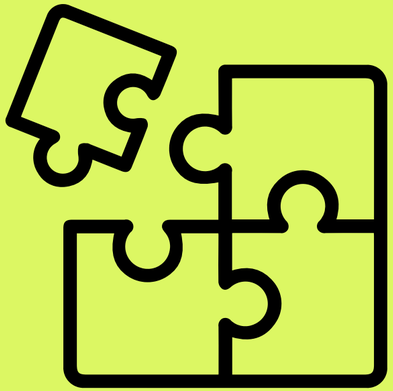
DataSet ค่อนข้างหาได้ยาก



จำนวน frame ของ video ไม่เท่ากัน

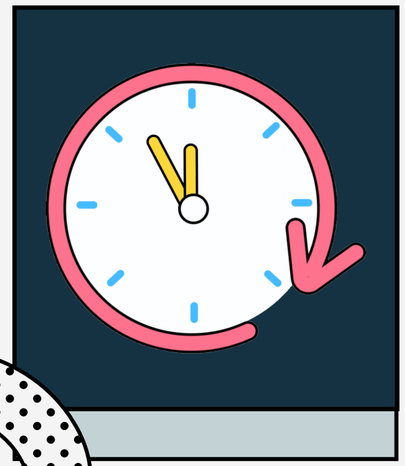
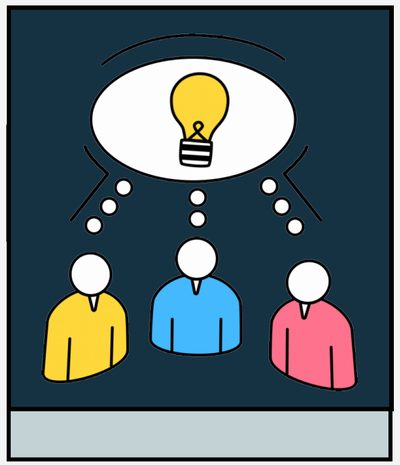
ทำ DataSet ขึ้นมาเอง

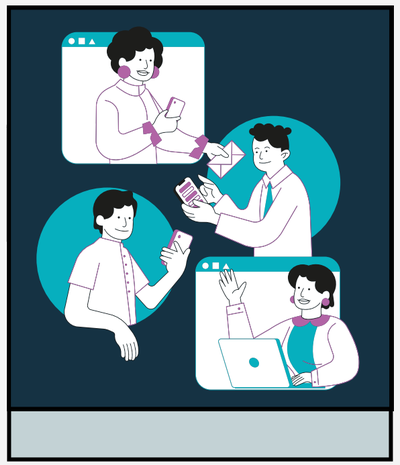
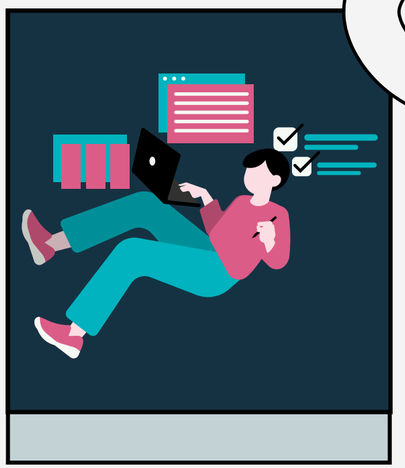




เขียนโค๊ดล็อกเฟรมให้เท่ากัน

PROJECT TEAM LESSON LEARN

Ideas from the project: Respect for each other is an inherent quality in teamwork.